Date: 5th April, 2026

ANRF AISEHack 2026

Finale presentation | Theme : Phase 2 Multiclass Flood Detection & Segmentation.

Team: who-knows

Devam Shah (solo)

Your salient contributions

f1

Modelling strategy

Physics-informed 15-channel expansion & U-Net++ architecture.

f2

Training

Boundary-aware Tri-Loss formulation and Stochastic Weight Averaging (SWA).

f3

Results + Actions

Ensemble 8-way Test Time Augmentation (TTA) and morphological noise refinement.

MATCHMAKING

Modelling - *Strategies*

01 **Physics-Informed Engineering**: Expanded raw 6-channel TIFFs to 15-channel tensors. Injected **NDWI**, **MNDWI**, and **AWEI** to suppress urban shadows and highlight water boundaries.

02 **Advanced Architecture**: Utilized **ADLinkNet** with **Xception, ResNet, EfficientNet** backbones and **U-Net++** with **EfficientNet, ResNet** backbone. Dense nested skip connections bridge the semantic gap between turbulent floodwater and static water bodies.

03 **Domain-Shift Proofing**: Patched ImageNet-pretrained convolutional stems to accept 15 channels. Early layers frozen to leverage generalized features while adapting to SAR-specific backscatter

Strengths: High flood recall; robust to shadow-induced false positives.

Weaknesses: Higher parameter count increases inference latency; sensitive to SAR speckle noise

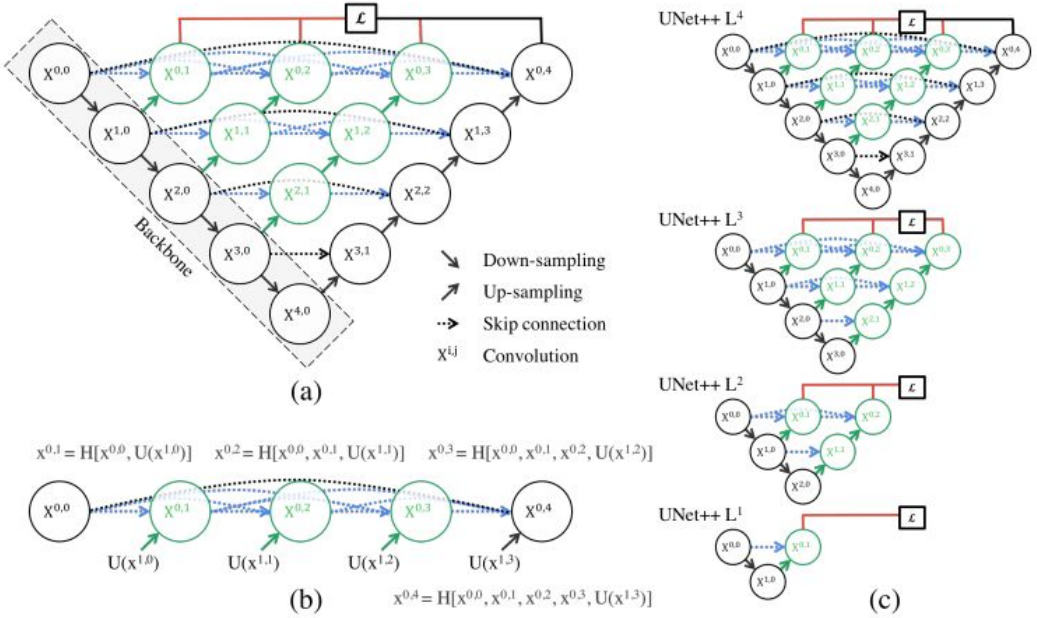
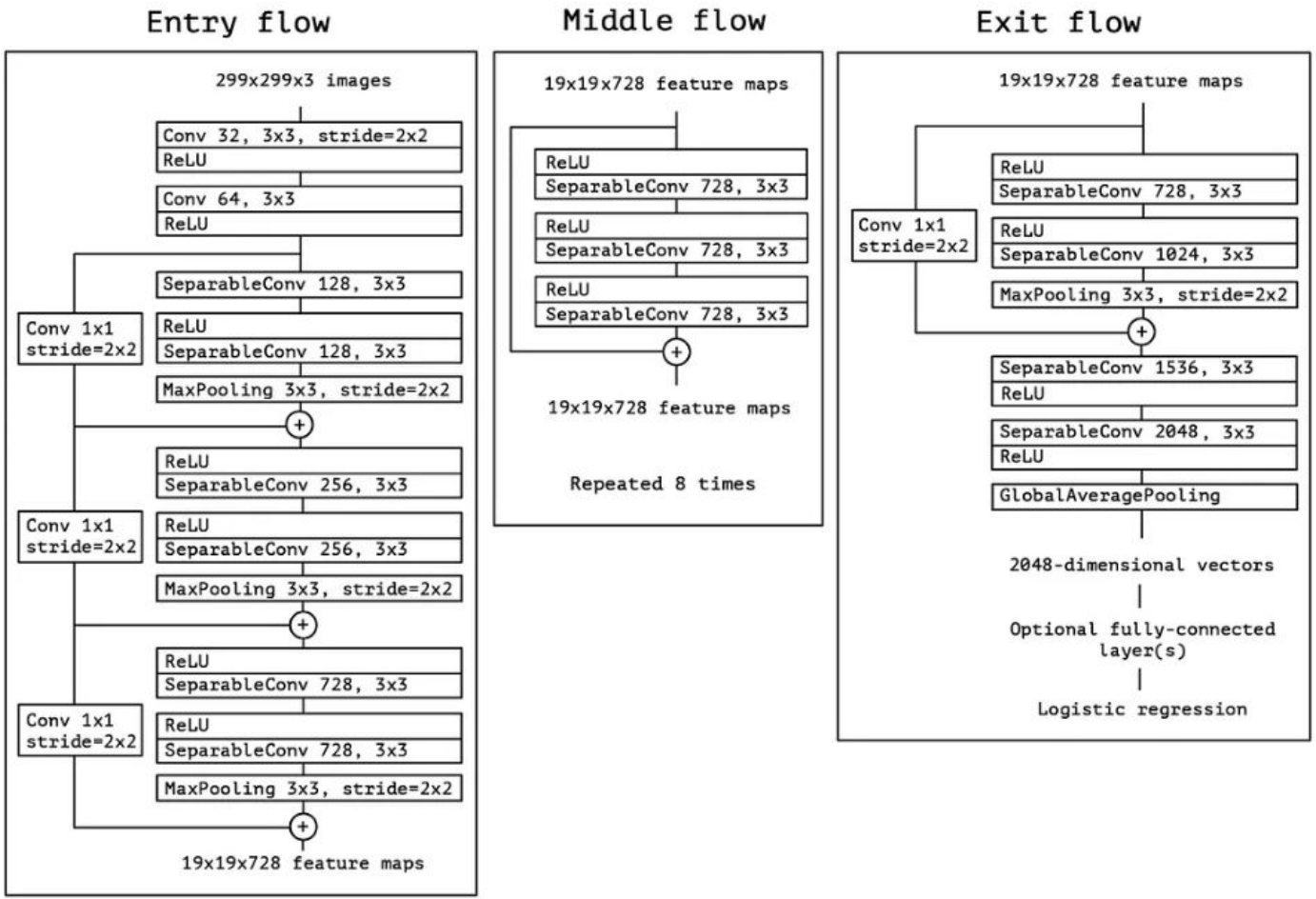


Fig. 1: (a) UNet++ consists of an encoder and decoder that are connected through a series of nested dense convolutional blocks. The main idea behind UNet++ is to bridge the semantic gap between the feature maps of the encoder and decoder prior to fusion. For example, the semantic gap between ($X^{0,0}$,$X^{1,3}$) is bridged using a dense convolution block with three convolution layers. In the graphical abstract, black indicates the original U-Net, green and blue show dense convolution blocks on the skip pathways, and red indicates deep supervision. Red, green, and blue components distinguish UNet++ from U-Net. (b) Detailed analysis of the first skip pathway of UNet++. (c) UNet++ can be pruned at inference time, if trained with deep supervision.

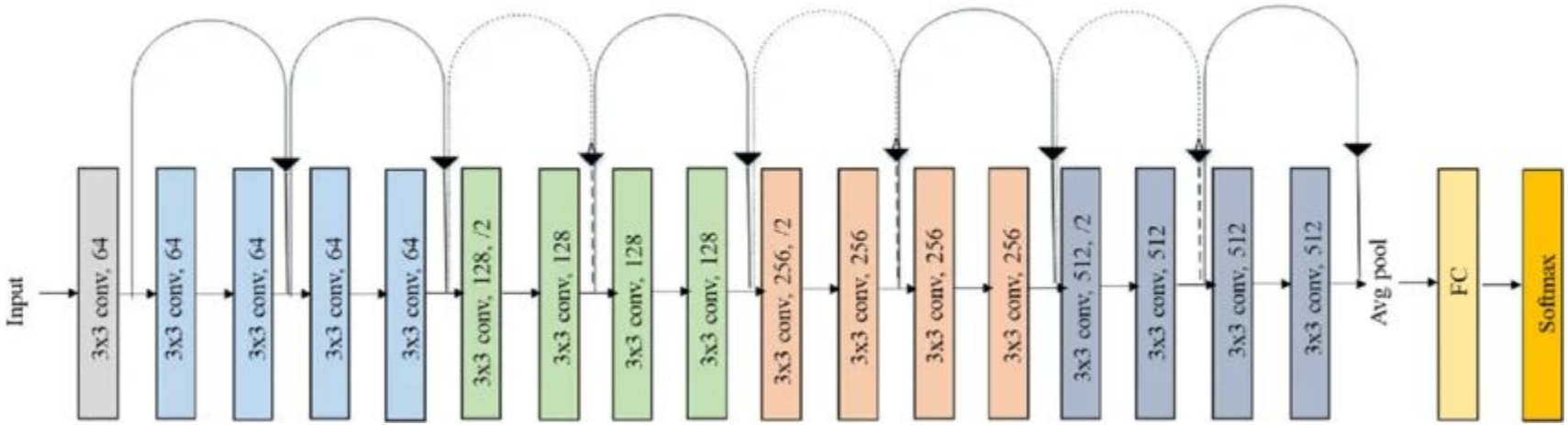
UNET++

https://arxiv.org/pdf/1807.10165

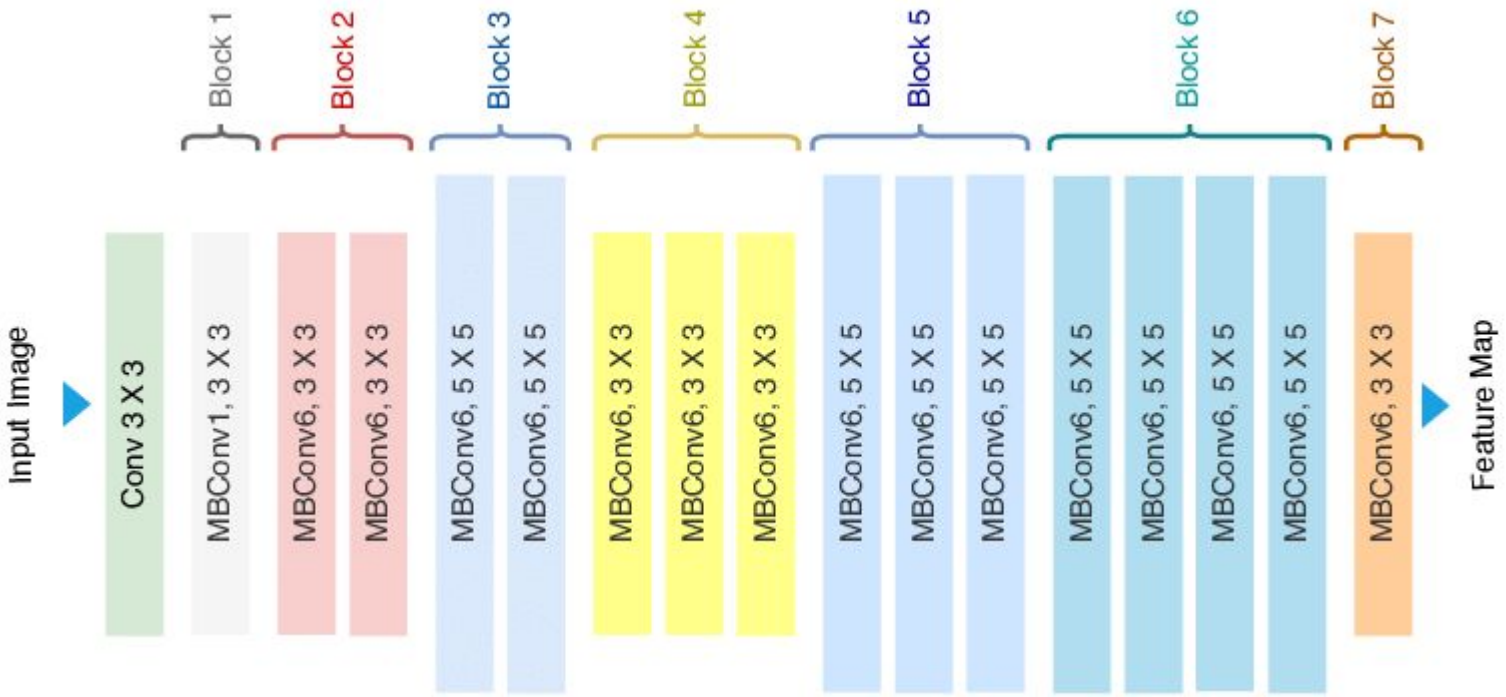


Xception

https://openaccess.thecvf.com/content_cvpr_2017/papers/Chollet_Xception_Deep_Learning_CVPR_2017_paper.pdf


Input
3x3 conv, 64
3x3 conv, 64
3x3 conv, 64
3x3 conv, 64
3x3 conv, 64
3x3 conv, 128, /2
3x3 conv, 128
3x3 conv, 128
3x3 conv, 128
3x3 conv, 256, /2
3x3 conv, 256
3x3 conv, 256
3x3 conv, 256
3x3 conv, 512, /2
3x3 conv, 512
3x3 conv, 512
3x3 conv, 512
Avg pool
FC
Softmax

Resnet



Efficientnet

REVIEW GROUNDING

Key Results – *Why , what and how of the workings?*

S1

What is working?

Architecture: ADLinkNet ResNet18 and U-Net++ EfficientNet-B5 | **Training**: Tri-Loss (Weighted Focal + Multiclass Tversky + Boundary Loss). | **Data**: 5-fold Cross-Validation with SWA

S2

How is it working? (Quantitative evals)

Public LB Score: 0.1539 (ResNet18 baseline). | **Internal mIoU**: ~0.60 across 3 classes. | **Model Particulars**: ~32M parameters; trained on **NVIDIA Tesla T4**; Inference time: <1.0s/sample.

S3

Why is it working? (Reasoning and analysis)

Paradigm Shift: Moved from Phase 1 binary classification to handling 3-class spectral overlap. | **Tversky Tuning**: Penalizing False Positives (beta=0.7) prevented the "over-prediction" of flood on permanent water.

MATCHMAKING

Key Results - *Why , what and how of the workings?*

01

Backbone Scaling: Xception and ResNet34 exhibited overfitting on the limited training split.
Pivots: Transitioned from heavy backbones back to a lighter ResNet18 for robustness, then finally to an optimized U-Net++ ensemble.

02

Backbone Comparison: Zhang et al. (2020) for backbone selection.
U-Net++: Zhou et al. (2018) for dense feature fusion.
Academic Base: Foundational knowledge from **(BSDA2001)** Course

03

Used **Claude 4.6 Sonnet** and **Gemini 2.5 Pro/Flash** to implement AD-LinkNet and iterate on annealed loss strategies.
Chain-of-Thought (CoT) focused on solving the "salt and pepper" RLE noise through post-processing logic.

MATCHMAKING

Final Refinement & Novelty

- **Novelty**: Introduction of a **Tri-Loss formulation** and SAR-specific multiplicative speckle noise augmentation.
- **Stabilization**: Implemented **8-way TTA** (combining rotations and flips) to reduce classification variance on edge pixels.
- **Contiguity Actions**: Applied morphological filtering (min-area 120 pixels) to clean isolated predictions before final RLE generation.
- **Final Output**: Domain-shift resistant model producing contiguous, RLE-encoded flood masks.